LAST NAME: _____

Practice Quiz 09C — MATH 2055 — Fall 2026

Practice for Quiz 09

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Let $f : A \to B$ be a function and let $Y, Z \subseteq B$. Prove
$$f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z).$$

   (b) A student claims that $\sum_{i=0}^{n} 2^i = 2^{n+1}$ for every integer $n \geq 0$ and writes: *Assume the formula holds at n. Adding 2^{n+1} gives 2^{n+2}, so it holds at $n+1$. Thus induction proves the claim.* Identify the gap by checking a base case, and state a corrected formula.